

Cyber Security
Class No 07
Date: 28 Sep 2024

CIA – Fundamental Concepts:

1. **C** – Confidentiality
 2. **I** – Integrity
 3. **A** – Accessibility
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Security Types:

- **Defensive Security:** Focuses on protecting systems.
 - **Offensive Security:** Involves testing security mechanisms.
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Penetration Testing (Ethical Hacking):

Identifying vulnerabilities through:

1. **Information Gathering:**
 - Types of testing:
 - Black Box
 - White Box
 - Gray Box
2. **Network Scanning:**
 - Scanning IP addresses.
 - Conducting detailed scans for specific IPs.
 - Identifying open ports for a target IP (ranges: 0–65535).
 - Sending probe packets (without data) to confirm open ports.
 - Common tools:
 - **Nmap/Zenmap**
 - **Hping2/Hpings**
 - **Masscan**

- **Red Teaming:** Advanced and unconventional testing.
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Discovery Scan:

- Use nmap -sn -PR <target IP> to perform a scan.
- Example for subnet: nmap -sn 192.168.10.0/24.
- A live host is detected if it appears as "host is up."

Example Command Output:

```
└─(root@kali)-[~]
```

```
└─# nmap -sn 192.168.0.101/24
```

Starting Nmap 7.94SVN (<https://nmap.org>) at 2024-10-04 13:32 EDT

Nmap scan report for 192.168.0.1

Host is up (0.0076s latency).

MAC Address: D8:32:14:63:32:E8 (Tenda Technology,Ltd.Dongguan branch)

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.13 seconds

Common Scanning Techniques:

1. Default TCP Scan:

- nmap -sT 192.168.10.100 (scans the first 1000 ports).
- For specific ports:
 - Single: nmap -sT 192.168.10.100 -p 80
 - Multiple: nmap -sT 192.168.10.100 -p 80,44,123
 - Range: nmap -sT 192.168.10.100 -p 80-1000
 - All ports: nmap -sT 192.168.10.100 -p-

2. UDP Scan:

- nmap -sU 192.168.10.100 -p 80

3. SYN Port Scan (No Data Transfer):

- nmap -sS 192.168.10.100 -p 80

4. Operating System Details:

- nmap 192.168.10.100 -p 80 -sv -O

5. Scan Speed Adjustment:

- Use -T0 to -T5 for scan speed (slowest to fastest).
- Default recommendation: -T4.

6. Firewall Evasion (Inverse Scan):

- Techniques:
 - -sX (Xmas Scan)
 - -Maimon Scan
 - Example: nmap 192.168.10.198 -sX -p 137,139,445.
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Domain Scan (Bypass IP Blocking):

- Use the decoy option: nmap 192.168.10.100 -D RND:10.
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Enumeration (Target IP Details):

1. SMTP Enumeration:

- Check port 25: nmap -p 25 192.168.10.100.
- Commands:
 - telnet 192.168.10.100 25
 - VRFY root or VRFY test.
- Use tools like Hunter.io to gather email service IPs.

2. Other Enumeration Commands:

- Create a user file:

nano users.txt

ctrl+X → Y → Enter

cat users.txt

1. smtp-user-enum -M VRFY -U users.txt -t 192.168.10.100 **Network File Systems:**

- Mount check:
 - showmount -e 192.168.10.100.

3. Identifying OS with Ping:

- TTL values:
 - **Linux:** 64 or 63.
 - **Windows:** 128 or 127.

For further commands, use `nmap --help`.